Lecture 20

Tuesday, 22 September 2026 8:23 AM

Recap: If f is Continuous & strictly ↑

then f is invertible.

$g = f^{-1}$

$$\frac{dg}{dy} = \frac{1}{\frac{df}{dx}}$$

$y = f(x)$
$x = g(y)$

Sequences:

$\{s_0, s_1, s_2, \dots\}$

$f(x) = x^2 - 2$

$$s_{n+1} = \frac{s_n^2 + 1}{2 s_n}$$

$s_0 = 3 \longrightarrow$ initial guess

$s_1 = \frac{s_0^2 + 1}{2 s_0} = \frac{5}{3} \approx$

$s_2 = \dots$

$s_3 = \dots$

$\vdots$ ↓

1.414...

Q: What is a sequence & when does it converge?

Sequence : $\mathbb{N} \longrightarrow \mathbb{R}$

Terms of sequence : $s_1, s_2, s_3, \dots$

Eg1: $s_{n+1} = \frac{s_n^2 + 2}{2 s_n}$, $s_0 = 3$

$\{3, \frac{5}{3}, \dots\}$

Eg2: $s_{n+1} = s_n + s_{n-1}$, $s_1 = 1$, $s_2 = 1$

$\{1, 1, 2, 3, 5, \dots\}$ Fibonacci sequence

When does a Sequence Converge?

Eg1: $s_n = \frac{1}{n}$ $\quad n \in \mathbb{N}$,

Sequence $= \left\{ 1, \frac{1}{2}, \frac{1}{3}, \dots \right\}$

↳ discrete form of

$f(x) = \frac{1}{x}$ $\quad x \in \mathbb{N}$

$s_n = \frac{1}{n} \longrightarrow 0$ as $n \to \infty$

Def: We say that $\lim_{n\to\infty} s_n = L$ OR $\{s_n\}$ converges to L if

For every $\varepsilon > 0$, $\exists$ a $N_\varepsilon \in \mathbb{N}$ s.t.

$|s_n - L| < \varepsilon$ whenever $n \geq N_\varepsilon$.

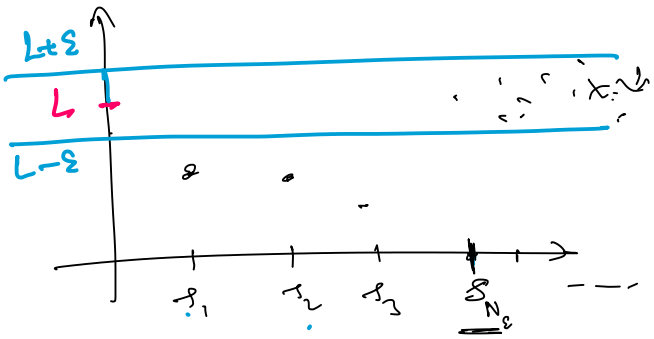
$\Downarrow$

$s_n \in (L-\varepsilon, L+\varepsilon)$

$|s_n - L| < \varepsilon$

$\Rightarrow -\varepsilon < s_n - L < \varepsilon$

$\Rightarrow L-\varepsilon < s_n < L+\varepsilon$

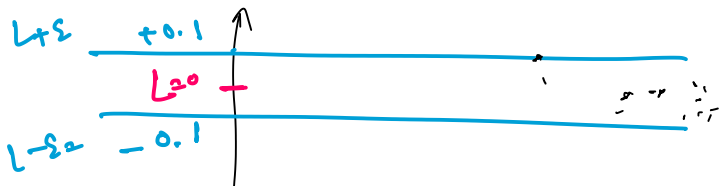


Eg1: $s_n = \frac{1}{n}$

$L = \lim_{n\to\infty} s_n = 0$ (Guess)

$\varepsilon = 0.1$

$s_1 = \frac{1}{1},$

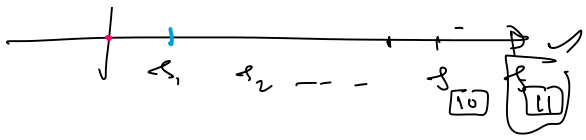


$f_2 = \frac{1}{2}$

$f_3 = \frac{1}{3}$

$\vdots$

$f_{10} = 0.1$



$\varepsilon = 0.01$

$f_n \in (0 - 0.01,\ 0 + 0.01)$ $\quad \forall n \geq$?

$\frac{1}{n} \in (-0.01,\ 0.01)$

$\Rightarrow \quad \frac{1}{n} < 0.01$

$\Rightarrow \quad n > \frac{1}{0.01} \Rightarrow n > 100$

$N_\varepsilon = 101$ or 102, ...

$\varepsilon = 0.01 \longrightarrow N_\varepsilon = 101$

In general for a given $\varepsilon > 0$,

$f_n \in (0 - \varepsilon,\ 0 + \varepsilon)$ whenever $n >$?

$\Rightarrow \quad \frac{1}{n} \in (-\varepsilon, \varepsilon)$

$\Rightarrow \quad \frac{1}{n} < \varepsilon$

$\Rightarrow \quad n > \frac{1}{\varepsilon}$ $\qquad N_\varepsilon = \left\lceil \frac{1}{\varepsilon} \right\rceil + 1$

For any choice of $\varepsilon > 0$, the recommended stage $N_\varepsilon = \left\lceil \frac{1}{\varepsilon} \right\rceil + 1$ such that

$f_n \in (0 - \varepsilon,\ 0 + \varepsilon) \quad \forall\ n \geq N_\varepsilon$

$\therefore$ By def, $f_n \longrightarrow 0$ as $n \to \infty$.

Eg 2: $f_n = \frac{1}{n^2}$

$L = \lim_{n \to \infty} f_n = 0.$ (Guess)

$\varepsilon = 0.1 \longrightarrow N_\varepsilon = 4$?

$s_n \in (0-0.1,\ 0+0.1)$ $\quad \forall n \geq \boxed{4}$

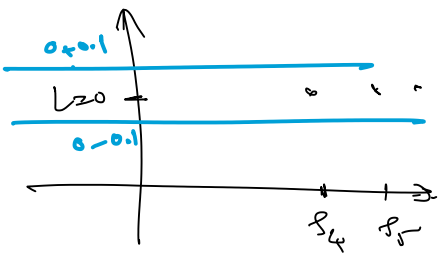
$\Rightarrow s_n < 0.1$

$\Rightarrow \frac{1}{n^2} < 0.1$

$\Rightarrow n^2 > 10$

$\Rightarrow n > \sqrt{10}$

$N_\varepsilon = N_{0.1} = \lceil \sqrt{10} \rceil = 4$



$\varepsilon = 0.01 \longrightarrow N_\varepsilon = 11$? check!

In general, for any $\varepsilon > 0$, the recommended stage is $N_\varepsilon = \lceil \sqrt{\frac{1}{\varepsilon}} \rceil + 1$ such that

$s_n \in (0-\varepsilon,\ 0+\varepsilon) \quad \forall\ n \geq N_\varepsilon = \left\lceil \sqrt{\frac{1}{\varepsilon}} \right\rceil + 1$

$\frac{1}{n^2} < \varepsilon \Rightarrow n^2 > \frac{1}{\varepsilon}$

$\Rightarrow n > \sqrt{\frac{1}{\varepsilon}}$

Eg 2: $s_n = e^{-n}$, $L = 0$ (Guess)

Let $\varepsilon > 0$ be given

We need to find a stage N_ε s.t.

$s_n \in (0-\varepsilon,\ 0+\varepsilon) \quad \forall\ n \geq \boxed{?}$

Target: $s_n < \varepsilon$

$\Rightarrow e^{-n} < \varepsilon$

$\Rightarrow \log_e(e^{-n}) < \log_e \varepsilon$

$\Rightarrow -n < \log_e \varepsilon \Rightarrow n > -\log_e \varepsilon$

$\Rightarrow n > \log_e\left(\frac{1}{\varepsilon}\right)$

$$N_\varepsilon = \left\lceil \log_e\left(\tfrac{1}{\varepsilon}\right) \right\rceil + 1$$

Ex: $\quad s_n = \dfrac{n}{n+1} \qquad L = 1$

For $\varepsilon > 0$, $\quad N_\varepsilon = ?$